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Optimizing saffron cormlet production through substrate composition nutrient concentration and irrigation management in soilless cultivation

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A two-year study has been conducted to optimize saffron cormlet production in a soilless cultivation system. Variations in the concentration of phosphate, boron, and irrigation events were assessed in the first year. Subsequently, after optimizing the substrate composition, the effects of nutrient solution volume and the concentration of nitrate, iron, and boron were investigated on the yield and weight of cormlets and leaves, photosynthetic activities, and productivity of nutrient solutions in the second year. Irrigation events in the first year significantly influenced cormlet growth, while phosphate and boron had no substantial effects. Moisture characteristics indicated an optimal substrate composition of 15% cocopeat, 15% cocochips, and 70% perlite. In the second year, increasing nutrient solution volume (225 ml/pot/week) and nitrate concentration up to 9800 µM significantly increased the weight of the leaf, total photosynthesis rate, and large-sized cormlets (> 8 g) to nearly 50% of total cormlets. Conversely, increasing iron concentration notably decreased the weight of total and large-sized cormlets. Boron concentration again showed no significant effect on the parameters. The highest nutrient solution productivity was achieved with a 150 (ml/pot/week) nutrient solution containing 9800 µM nitrogen and 25 µM iron. These findings underscore the importance of effective irrigation and nutrition management in enhancing the quantity and quality of cormlet production, potentially boosting perennial saffron yield in the following years.

Keywords Saffron cormlet, Soilless, Photosynthesis, Irrigation, Nitrate, Iron

Saffron (*Crocus sativus* L.) is a low-volume high-value crop whose dried stigma is one of the most precious medicinal spices in the world[1]. It is used as a natural color and flavoring agent in the food industry and as a fragrance and coloring agent in perfumes and the textile industry. It is also used in producing numerous anti-cancer, anti-Alzheimer, and anti-depressant drugs[2].

Saffron can, to some extent, tolerate harsh environmental conditions such as severe temperature fluctuations during the day and throughout its growth period, as well as limited access to water. It thrives in arid and semi-arid regions with an average rainfall of about 100–300 mm/yr, supplemented by occasional irrigation with well water (averaging 3000–3600 m^3/ha/yr.)[3]. However, global climate changes over the past few decades have exacerbated the challenges of traditional saffron cultivation. These changes have led to a decrease of approximately 35% in saffron stigma yield during this period[4,5]. This decline can be attributed to rising temperature and a concurrent reduction in average annual rainfall, both of which directly impact saffron growth and development, while also indirectly diminishing soil organic matter and fertility.

Saffron, an agronomical perennial geophyte, is a sterile triploid plant that does not produce seeds. Instead, it is propagated annually by its corms[6]. In geophytes, the production of cormlets (daughter corms) is influenced by the quality of mother corms[7], which somewhat depends on the availability of nutrients during the plant's growth and development stages[8]. Saffron flowers before the leaves emerge. Within flowering period, the nutritional

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compounds stored in the saffron corms are used for flowering and developing leaves and roots. Any remaining nutrients in the (mother) corms are mobilized to newly born cormlets. Therefore, heavier corms lead to higher productivity[9]. After leaves development, the photosynthesis process will support the growth of cormlets from early winter to spring[10]. Therefore, the photosynthesis rate is a key factor that affects the production of larger-sized cormlets and more flowers and stigmas. The availability of light, temperature, carbon dioxide, water, and nutrients affect the photosynthesis rate[11]. The weight of corms also influences the size of flowers and stigmas[12], the heavier the corms, the more productive.

In saffron production, irrigation management plays a key role in nutrient uptake, followed by photosynthetic rate and cormlet formation[13]. Water deficit and/or improper irrigation management are major causes of saffron yield reduction[14]. Therefore, optimal control of environmental conditions is the most effective strategy to increase or maintain crop yields, especially under harsh environmental conditions. In combination with greenhouses, hydroponic cultivation is considered the most efficient system due to the maximum control over water and nutrient usage[15]. This system is expected to result in a higher saffron yield[16], and potentially increase the main components of saffron, including crocin, picrocrocin, and safranal, which are indicators of the saffron quality[17]. The optimization of factors affecting growth, such as substrate composition and irrigation events, is crucial for successfully running a hydroponic cultivation system to prevent plant stress[18]. The composition of the substrate influences water, nutrients, and oxygen availability in the root environment[19]. Commercial substrates are good candidates to be used for saffron production[17]. For instance, it has been shown that cultivating saffron in vermiculite and perlite positively affects yield and the number of produced cormlets[20] and cultivation in cocopeat and perlite increases the dry weight of cormlets[21].

Cocopeat, which is widely used as a substrate material, has a high water-holding capacity, resulting in limited aeration in the root environment. Depending on its preparation process, the amount of air permeability and water holding capacity may vary from 11 to 53% and 50 to 81%, respectively[22]. To improve aeration in the root environment and provide a proper capacity for water and nutrient uptake, incorporating coarser materials into cocopeat can be an effective solution[23]. In addition, maintaining a balanced ionic composition is crucial for optimal plant growth and nutrient acquisition. Nitrogen (N) is the most important macronutrient for plant growth as it plays a key role in the process of photosynthesis[24]. The effect of a chemical and/or organic source of nitrogen on saffron flowering and stigma yield has been highlighted in the last four decades[25]. Insufficient nitrogen can lead to reduced saffron leaf growth and the production of smaller corms[26]. Iron (Fe) is another essential nutrient that plays a fundamental role in various plant metabolism processes, such as photosynthesis, respiration, nitrogen fixation, and chloroplast maintenance. Around 80% of plants' Fe content is found in photosynthetic cells[27], with 12 Fe atoms in photosystem I (PSI), two or three in PSII, five in cytochrome complexes, and two in ferredoxin molecules. Besides, Fe is a key factor in the electron transport chain in the photosynthesis process[28] and is involved in Fe-S clusters of chloroplasts[29]. Although iron availability can significantly impact the photosynthesis and productivity of plants[30], high iron concentrations may significantly decrease the growth of plants[31]. In a few documents, the positive effect of iron by foliar[32] and soil application[33] has been reported on different saffron traits, however, the optimum Fe concentration in the hydroponic nutrient solution is unknown. Therefore, a proper supply of all nutrients, including N and Fe, is crucial for metabolic processes, which would increase saffron cormlets and flower yield[34].

Currently, there is a lack of information available for setting up a soilless cultivation system for saffron. Research on the hydroponic cultivation of saffron mainly focuses on optimizing growth-related factors based on the available conditions and facilities. Given the significance of this issue, the first aim of this study was to determine an appropriate irrigation water volume based on the composition of the substrate used in the cultivation process. On the other hand, while some studies have investigated the effect of electrical conductivity[35], temperature, and oxygen concentration of nutrient solutions[36] on saffron yield, the effect of ionic composition of the nutrient solution has rarely been documented. Thus, the second aim of this study was to investigate the effect of different concentrations of nitrate, iron, boron, and phosphate in the nutrient solution on the production of saffron cormlets individually. This information provides basic knowledge for the soilless production of saffron stigma and cormlet, which is becoming increasingly important due to the decreasing availability of arable land and water scarcity, by providing insight into manipulating the ionic composition of a more proper nutrient solution for saffron.

Results

The first-year experiment

The effect of irrigation events

Table 1 shows the results of evaluating the impact of different irrigation events (daily, twice weekly, and weekly) on the weight of cormlets produced. The twice-weekly irrigation significantly increased the yield of both total cormlets and large-sized (5 to 9 g) cormlets. Conversely, the daily irrigation resulted in the highest percentage of cormlet loss (44%), leading to a decrease in the yield of both total and large-sized cormlets. Although the weekly irrigation reduced cormlet loss, it led to a significant reduction in cormlet yield, exceeding 50%. In addition, in all treatments, the saffron roots became woody (Fig. 1a) and the corm to root ratio decreased, probably due to the composition of the substrate used (70:30, cocopeat to perlite), in contrast to the normal roots achieved through optimized substrate composition in the second year (Fig. 1b). Furthermore, the effect of phosphate and boron treatments on cormlet yield was insignificant likely due to the loss of a large number of cormlets, especially in the daily irrigation event.

On the other hand, the initial observations before the completion of the saffron plant's growth cycle indicated that the irrigation treatment has exerted a discernible influence on the physiological aspects of saffron cormlet development, concurrently resulting in a shorter growth period. Figure 1c demonstrates that increasing the irrigation interval from daily to weekly led to an accelerated yellowing of saffron plant leaves, occurring

Treatments	Total cormlet g/corm	Large cormlet g/corm	Lost cormlet %
Daily	2.40 ± 0.28^b	1.37 ± 0.43^b	44
Twice weekly	4.25 ± 0.14^a	2.30 ± 0.45^a	15
Weekly	2.28 ± 0.12^b	1.09 ± 0.54^b	7

Table 1. The effect of irrigation events (daily, twice weekly, and weekly) on the total cormlet weight, large-sized cormlet weight and the percentage of lost cormlet. Values are the mean of 6 replicates (two pots, three corms in each). The means followed by at least oneletter in common are not significantly different at the 5% level of probability.

notably earlier (in mid-February) than their anticipated physiological timing (in April), and growth cessation. Specifically, saffron plants receiving irrigation twice a week showed leaf discoloration in the latter half of March, while those under daily irrigation experienced leaf discoloration in the latter half of April.

Optimizing substrate composition based on moisture characteristics
Following the first year's experimental results, the moisture properties of various substrates composed of a mixture of cocopeat and perlite were evaluated. The results are presented in Table 2. Substrates 1 to 4 show how the size of perlite particles and their combination with cocopeat affect the overall porosity, as well as the percentage of water and air in the substrates. The first three rows demonstrate that different substrate compositions can yield the same total porosity but with various water and air content. The fourth raw highlights that utilizing coarse perlite increases the overall porosity of the substrate. However, due to the large pore size, the water-holding capacity decreases. This implies that the distribution of fine and coarse pores should also be taken into consideration, in addition to the total porosity.

Substrate 5 represents the composition used in the first year, consisting of 70% cocopeat and 30% perlite. Under these conditions, the saturation limit is 70%, the water holding capacity is 39%, and the gas phase is 31%. The corms and roots grown in this substrate showed that the water content of such a substrate is higher than what saffron requires. These conditions led to decay and microbial contamination of the corms, and changes in the root structure. This indicates the need to reduce the percentage of cocopeat, for instance by using a ratio of 30% cocopeat to 70% perlite, as presented in substrate 6. This composition provides more porosity and less moisture, although the moisture may not be sufficient for normal saffron growth. In the last combination (substrate 7), half of the cocopeat was replaced with cocochips, resulting in larger particles in the organic part. This composition increased the moisture content and improved air circulation in the substrate.

The second-year experiment

The number and weight of leaves and cormlets
Table 3 shows that the change in the concentration of nitrate and iron in the nutrient solutions and the change in the volume of nutrient solution had no significant effect on the number of leaves. However, these treatments significantly affected the weight and leaf area of leaves. The highest volume of nutrient solution (V3 = 225 mL/pot/w) resulted in a 29% increase in leaf weight compared to the lowest volume (V1 = 75 mL/pot/w). Moreover, the highest iron concentration (Fe3 = 30 µM) significantly enhanced leaf weight. While elevating nitrate concentration up to N3 (9800 µM) led to an increase in leaf weight, further increases in nitrate concentration (N4 = 10100 µM) had a diminishing effect on this parameter.

The change in nutrient solution volume and iron concentration did not affect the total number of cormlets. Similarly, the change in nitrate concentration did not reveal a consistent pattern in influencing this parameter. However, increasing the volume of the nutrient solution increased the total weight of cormlets (Fig. 2a). The highest volume of nutrient solution (V3 = 225 mL/pot/w) produced the maximum cormlet weight (16.11 ± 1.82 g/corm), which was 66% higher than the V1 = 75 mL/pot/w treatment. Increasing nitrate concentration up to N3 = 9800 µM resulted in a substantial increase in both total and large-sized cormlet weights (Fig. 2b). In line with the findings for leaf weight, the maximum cormlet weight (16.36 ± 2.44 g/corm) was achieved at the third level of nitrate (N3 = 9800 µM), indicating a notable 23% increase compared to the N1 = 8100 µM treatment.

On the other hand, a negative effect on the weight of cormlets was found when the nitrate concentration increased to N4 = 10,100 µM. This resulted in a 25% decrease in total weight and a 52% decrease in the weight of larger-sized cormlets when compared to N3 = 9800 µM (Fig. 2b). Similar effects were observed in the Fe treatments, where an increase in the concentration from Fe1 = 20 to Fe3 = 30 µM resulted in a 35% decrease in the total weight of cormlets (Fig. 2c). Conversely, increasing the volume of nutrient solution significantly increased the weight of large-sized cormlets (Fig. 2a). The highest volume (V3 = 225 mL/pot/w) led to a 3.2-fold increase in the weight of large-sized cormlets. Boron, however, did not have any significant effect on any of the leaf and cormlet growth parameters, as shown in Table 3; Fig. 2d.

The experimental data reveals a significant and positive correlation ($R = 0.88$, $p \leq 0.001$) between the weight of cormlets and the weight of dried leaves, regardless of the type of treatment (Fig. S1). This correlation is observed in treatments involving both nutrient solution volume and nitrate concentration (Fig. 3a and b). However, despite an increase in leaf weight with higher iron concentration (Table 3), a strong negative relationship is found between the weight of leaves and the weight of both total cormlets and large-sized cormlets (Fig. 3c).

Fig. 1. (**a**) A saffron corm with woody roots and signs of microbial decay on the corms, most likely due to irrigation treatment in the first year, (**b**) two saffron corms with normally grown roots after optimizing substrate composition in the second year, and (**c**) the effect of different irrigation events on the duration of saffron leaf growth: daily irrigation (green leaves) versus weekly irrigation (yellow leaves), applied in the first year.

Substrate	Organic material		Perlite, in mm			Moisture%		Porosity %
	Cocopeat	Cocochips	1–3	3–5	7–10	Saturation	WHC	
1	0	0	0	100	0	64	21	43
2	10	0	0	90	0	64	35	29
3	10	0	90	0	0	63	32	31
4	10	0	30	0	70	78	23	54
5	70	0	0	30	0	70	39	31
6	30	0	0	70	0	63	24	39
7	15	15	0	70	0	78	43	35

Table 2. The effect of the type (cocopeat, cocochips, and perlite in three sizes) and ratio of substrate components on the percentage of saturation moisture, water holding capacity (WHC), and porosity.

Treatments	Number of leaves No/corm	Weight of leaves g/corm	Leaf area cm^2/corm	Number of cormlets No/corm	Weight of cormlets g/corm
N1	5.86 ± 0.77^{a}	0.27 ± 0.06^{c}	26.94 ± 1.24^{b}	5.17 ± 0.75^{a}	12.84 ± 1.55^{b}
N2	6.51 ± 0.29^{a}	0.31 ± 0.03^{b}	28.96 ± 1.96^{b}	4.51 ± 0.44^{ab}	13.57 ± 1.08^{b}
N3	5.86 ± 0.81^{a}	0.34 ± 0.05^{a}	43.95 ± 0.65^{a}	3.47 ± 0.93^{b}	16.36 ± 1.66^{a}
N4	5.82 ± 0.28^{a}	0.25 ± 0.03^{c}	22.04 ± 0.91^{c}	4.59 ± 0.79^{a}	12.98 ± 1.98^{b}
Fe1	5.74 ± 1.19^{a}	0.29 ± 0.07^{b}	41.07 ± 0.51^{a}	5.07 ± 1.20^{a}	15.10 ± 0.71^{a}
Fe2	6.51 ± 0.29^{a}	0.31 ± 0.03^{ab}	28.96 ± 1.96^{c}	4.51 ± 0.44^{a}	13.57 ± 1.08^{b}
Fe3	6.43 ± 0.73^{a}	0.34 ± 0.07^{a}	36.02 ± 1.02^{b}	4.40 ± 1.07^{a}	9.76 ± 0.90^{b}
B1	6.49 ± 0.73^{a}	0.35 ± 0.04^{a}	29.54 ± 2.18^{a}	4.35 ± 0.58^{a}	14.25 ± 0.86^{a}
B2	6.51 ± 0.29^{a}	0.31 ± 0.03^{a}	28.96 ± 1.96^{a}	4.51 ± 0.44^{a}	13.57 ± 1.08^{a}
B3	5.27 ± 0.16^{a}	0.26 ± 0.02^{a}	26.13 ± 0.86^{b}	4.67 ± 0.88^{a}	12.28 ± 0.62^{a}
B4	6.18 ± 0.80^{a}	0.30 ± 0.04^{a}	28.51 ± 2.47^{a}	4.07 ± 0.45^{a}	11.85 ± 1.11^{a}
V1	5.67 ± 1.92^{a}	0.30 ± 0.12^{b}	24.41 ± 1.04^{b}	3.60 ± 1.44^{a}	9.70 ± 2.89^{b}
V2	6.51 ± 0.29^{a}	0.31 ± 0.03^{b}	28.96 ± 1.96^{b}	4.51 ± 0.44^{a}	13.57 ± 1.08^{ab}
V3	6.44 ± 1.00^{a}	0.39 ± 0.09^{a}	48.35 ± 1.16^{a}	4.57 ± 1.05^{a}	16.11 ± 1.82^{a}

Table 3. The effect of nitrate (N1 = 8100, N2 = 9300, N3 = 9800, and N4 = 10100 µM), iron (Fe1 = 20, Fe2 = 25, and Fe3 = 30 µM), and boron (B1 = 18, B2 = 25, B3 = 32, and B4 = 46 µM) concentrations and nutrient solution volume (V1 = 75, V2 = 150, and V3 = 225 mL/pot/w) on the number and weight of both leaves and cormlets. Values are the mean of 15 replicates (five pots, three corms in each). The means in each column followed by at least one letter in common are not significantly different at the 5% level of probability.

Net and total photosynthetic rate
The treatments significantly affected the net and total photosynthetic rate (Figs. 4 and 5). Increasing the volume of nutrient solution from V1 = 75 to V3 = 225 mL/pot/w led to a 2.7-fold increase in both the net and total photosynthetic rates. While increasing the volume of nutrient solution from V2 = 150 to V3 = 225 mL/pot/w had no significant effect on the net photosynthetic rate (Figs. 4a and 5a). The maximum total photosynthetic rate (64 µmol $CO_2/m^2/s$) was achieved at the highest nutrient solution volume (V3 = 225 mL/pot/w).

Increasing the nitrate concentration from N1 = 8100 to N4 = 10,100 µM caused a significant increase in the net photosynthetic rate by 59% (Fig. 4b), while there were no significant net photosynthetic responses to increasing nitrate concentrations from N2 = 9300 to N4 = 10,100 µM. A similar increasing trend was observed for the total photosynthetic rate, with a 2.39-fold increase by increasing the concentration of nitrate from N1 = 8100 to N3 = 9800 µM (Fig. 5b). However, further increasing nitrate concentration to N4 = 10,100 µM caused a sharp reduction in total photosynthetic rate, leading to almost 1.1-fold decrease compared to the previous level.

Furthermore, increasing Fe concentration from Fe1 = 20 to Fe2 = 25 µM increased both the net and total photosynthetic rate. However, increasing iron concentration to Fe3 = 30 µM significantly decreased the net photosynthetic rate by 30%, while the total photosynthetic rate increased slightly.

The effect of photosynthetic rate on the weight of cormlets
The effect of the net and total photosynthesis rates on the weight of cormlets is shown in Fig. S2 as a function of experimental treatments. Both the total and large-sized cormlets showed a strong positive correlation with the net and total photosynthetic rate in relation to nutrient solution volume. In nitrate concentration treatment a positive and strong relation is observed with total photosynthesis rate while there is no correlation between net photosynthesis rate and cormlet weights (Fig. S2a, b). Conversely, in iron concentration treatment cormlets' weight negatively correlated with the total photosynthetic rate but there is no correlation with net photosynthesis rate (Fig. S2c).

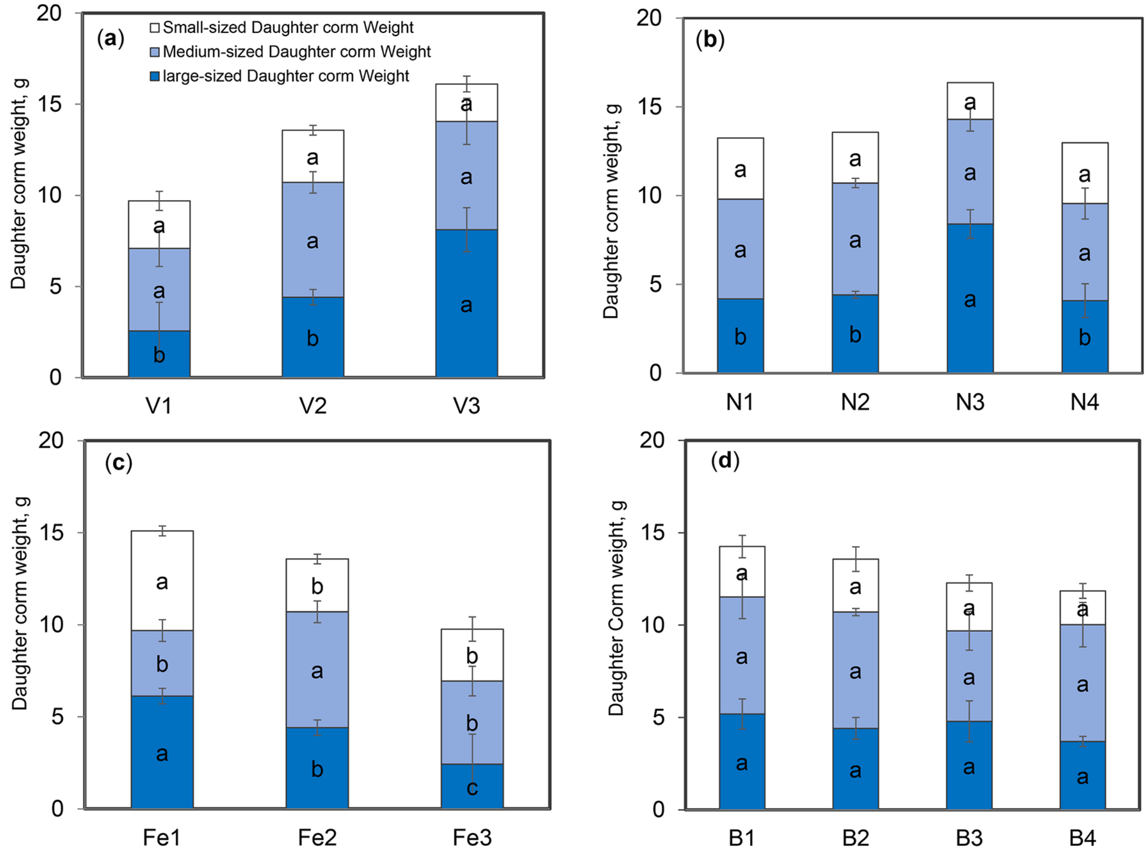

Fig. 2. The effect of (**a**) nutrient solution volume, (**b**) nitrate concentration, (**c**) iron, and (**d**) boron concentration on the mean weights of cormlets classified in three different sizes; small (< 4 g), medium (4–8 g) and large (> 8 g) sized cormlets. Values are means ± standard error ($n = 15$). The means followed by the same letter are not significantly different from each other, using LSD's multiple tests ($P < 0.05$).

Transpiration rate and water use efficiency
The effect of experimental treatments on the transpiration rate (TR) and instantaneous water use efficiency (WUE) are shown in Figs. S3 and 6. Increasing nutrient solution volume from V1 = 75 to V3 = 225 mL/pot/w substantially increased (2.16-fold) the transpiration rate (Fig. S3a) and slightly increased (12%) the WUE (Fig. 6a). While nitrate concentration significantly ($p < 0.05$) increased the transpiration rate, it had no statistically robust effect on WUE (Figs. S3b and 6b). Although LSD indicated some pairwise differences, the overall ANOVA revealed no significant impact of nitrate treatments on WUE. Similarly, increasing iron concentration increased the transpiration rate, but decreased WUE (Figs. S3c and 6c).

Nutrient solution productivity
Considering the economic importance of large-sized corms (weighing above 8 g) in saffron flowering, the nutrient solution productivity (NSP) was determined by calculating the weight of large-sized cormlets (g) per unit of applied nutrient solution (L). Figure 7 shows the NSP of the experimental treatments for the production of large-sized cormlets. Increasing the volume of nutrient solution from V1 = 75 to V2 = 150 mL/pot/w considerably decreased (47%) NSP. However, no significant ($p < 0.05$) difference was found between the lowest and highest nutrient solution levels, i.e., V1 = 75 and V3 = 225 mL/pot/w, (Fig. 7a). The NSP was greatly influenced by an increase in nitrate concentration, with a notable increase observed up to the N3 = 9800 µM (Fig. 7b). Consequently, the NSP at the N3 = 9800 µM treatment was 1.66 times higher than that at the N1 = 8100 µM treatment. However, further increasing the nitrate concentration to N4 = 10,100 µM resulted in a 63% reduction in NSP. Conversely, increasing the iron concentration in the nutrient solution negatively affected the NSP, causing a 50% decrease in the production of large-sized cormlets per unit of applied nutrient solution (Fig. 7c).

Nitrogen and iron accumulation in cormlets
The concentration of nitrogen and iron in the cormlets produced was increased significantly ($p < 0.05$) when the nitrate concentration in the nutrient solutions was increased up to N3 = 9800 µM (Table S6). The highest concentrations of nitrogen (26.6 mg/g of corm) and iron (44.6 µg/g of corm) were achieved with the N3 = 9800 µM treatment, which were 1.54 and 19.3 times higher than their concentration in the N1 = 8100 µM treatment, respectively. However, further increasing the nitrogen concentration to N4 = 10,100 µM resulted in decreasing

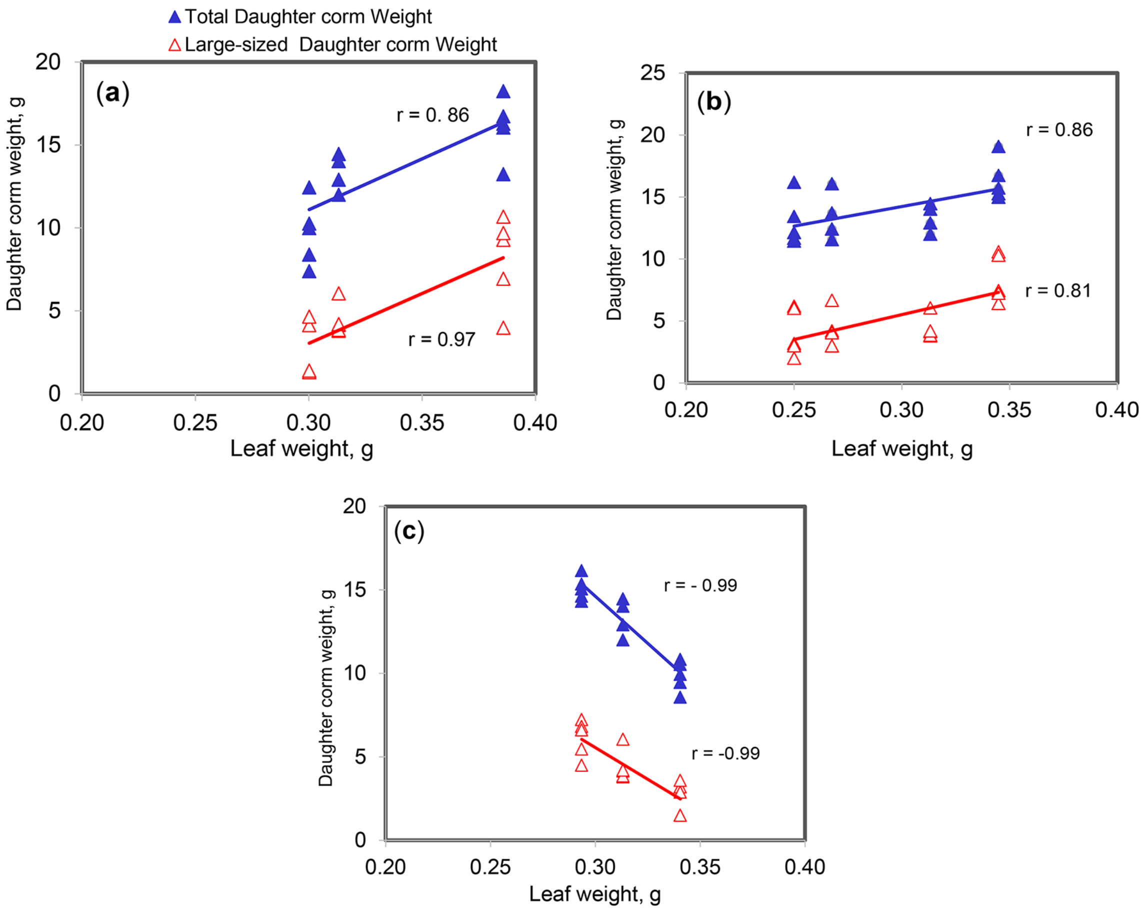

Fig. 3. The relationship between the weight of leaves and the weight of both total cormlets (circles) and large-sized cormlets (triangles) under different levels of (**a**) nutrient solution volume, (**b**) nitrate concentration, and (**c**) iron concentration.

the concentration of nitrogen and iron in the produced cormlets. The concentrations of nitrogen and iron in Fe1 = 20 µM were 1.5 and 1.36 times higher than their concentration in Fe3 = 30 µM, respectively.

Discussion

The first year's findings revealed that regulating moisture and air levels in the root environment plays a crucial role in adjusting the system of irrigation and nutrition for saffron. Consequently, in addition to exploring the effects of phosphate and boron on saffron growth, the first year's study also focused on examining the substrate's composition and its impact on the roots' ability to access water, nutrients, and air, which was considered a prerequisite.

During the first year of cultivation, it was observed that the growth and development of saffron corms and roots are influenced by the composition of the substrate and the irrigation events. This underscores the importance of adjusting the type and ratio of substrate components to improve the performance of the saffron plant. An appropriate ratio of cocopeat to perlite can regulate the substrate's level of porosity, saturated moisture, and water-holding capacity (WHC). The first-year response of saffron to irrigation and nutrition system and the substrate characterization (Table 2) indicated that saffron requires a well aerated substrate, which can be regulated by using a substrate consists of low and relatively coarse organic matter fraction. The results also showed that while both perlite and cocopeat can retain a high amount of moisture, their water availability to saffron roots may differ. It appears that the moisture of cocopeat can directly contact the roots, whereas the moisture stored within the tiny pores of perlite is less accessible. Therefore, adding cocochips to the substrate increases the total porosity, water holding capacity, and gas circulation, leading to a more even moisture distribution among the substrate's pores. The data presented in Table 2 supports these findings.

In the second year of experimentation, the effects of nutrient solution volume and the concentration of nitrate, iron, and boron on the production of saffron cormlets were investigated. The number of leaves was not affected by the experimental treatments; however, an increase in the weight of leaves was observed, which could be attributed to an increase in leaf area and/or thickness. The size of the planted corms was found to be the most important factor influencing the number of leaves, which were almost the same in our study. It was observed that an increase in the number of sprouts per corm led to a higher number of leaves in larger corms[10]. In addition, the growth and weight of leaves can be limited by irrigation and nitrate deficiency[37].

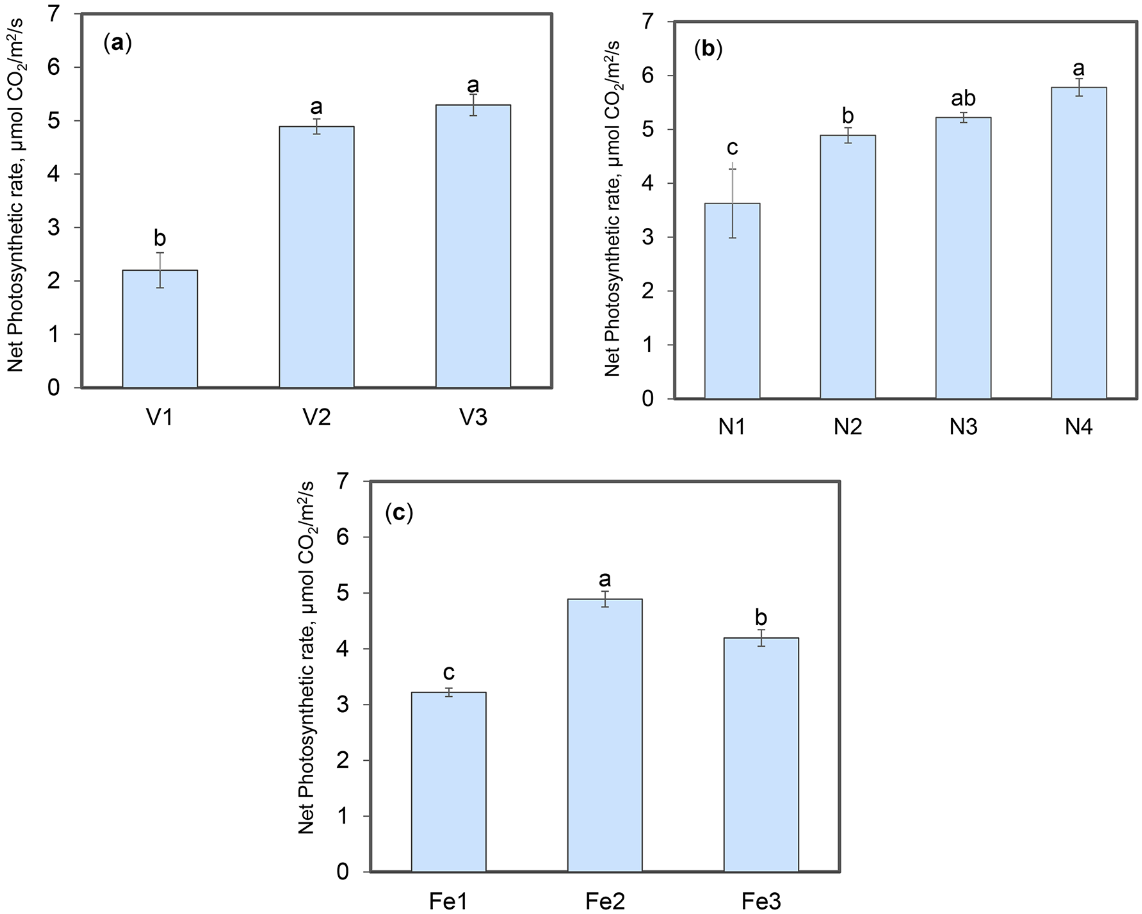

Fig. 4. The effect of (**a**) nutrient solution volume, (**b**) nitrate concentration, and (**c**) iron concentration on the mean net photosynthetic rate. Values are means ± standard error (n = 15). The means followed by the same letter are not significantly different from each other, using LSD's multiple tests (P < 0.05).

The photosynthetic rate of plants is considered the primary metabolic process determining the production of saffron cormlets, accounting for 87–91% of total corm formation[10]. Our results indicate that total photosynthetic rates were directly (net photosynthetic rate) and indirectly (leaf dry weight) influenced by the volume of nutrient solution and the chemical composition of the solutions, especially the concentration of nitrate and iron ions. Increasing the nutrient solution volume, as well as the nitrate and iron concentration, significantly increased the weight of leaves, indicating an improvement in leaf photosynthetic rate. The increase in cormlets weight was affected more by the total photosynthesis rate than net photosynthesis rate.

Furthermore, the growth response of saffron plants, which includes both leaf and corm weights, was highly influenced by the supply of nitrate. The optimal level of nitrate supply for saffron growth was 9800 µM (N3). However, when the nitrate supply exceeded 9800 µM (N3), a substantial decrease in plant growth occurred (Table 3). The total photosynthetic rates were also decreased when the nitrate concentration was increased up to 10,100 µM (N4). At this nitrate concentration, the reduction in leaf weight coincided with a decrease in leaf area, which significantly contributed to the overall photosynthesis rate. These findings suggest that there is a complex interplay of factors that influence photosynthesis and growth in this particular context.

Our findings indicate that, under our experimental conditions, the application of elevated nitrate concentrations (exceeding 9800 µM, N3) had a detrimental effect on saffron plant growth. Recent studies suggest that elevated nitrate accumulation leads to the generation of nitrite, which subsequently gets converted into nitric oxide (NO) within plants. Notably, nitrate reductase (NR) catalyzes NO and O^{2-} to form peroxynitrite ($ONOO^-$), a highly toxic compound for plants[38]. Although the nitrate concentration in our study were in adequate range[39] and did not surpass toxic levels, it is noteworthy there are no reports demonstrating the threshold toxic level of nitrate concentrations for saffron cultivation under hydroponic conditions.

Increasing the volume of nutrient solution led to a significant rise in net photosynthetic rates, which is consistent with previous observations on other plants[40]. This remarkable increase in net photosynthetic and transpiration values due to an increase in water volume suggests the importance of water availability on photosynthetic activity through stomatal regulation in saffron. Stomata closure and the consequent reduction in CO_2 uptake would decrease the photosynthetic rate of leaves[41]. Therefore, at higher water levels and a nitrate concentration of 9800 µM (N3), there were higher CO_2 exchange rates[42], stomata conductivity, transpiration, and net photosynthetic rate[43], which resulted in the production of large-sized cormlets, accounting for 50% of

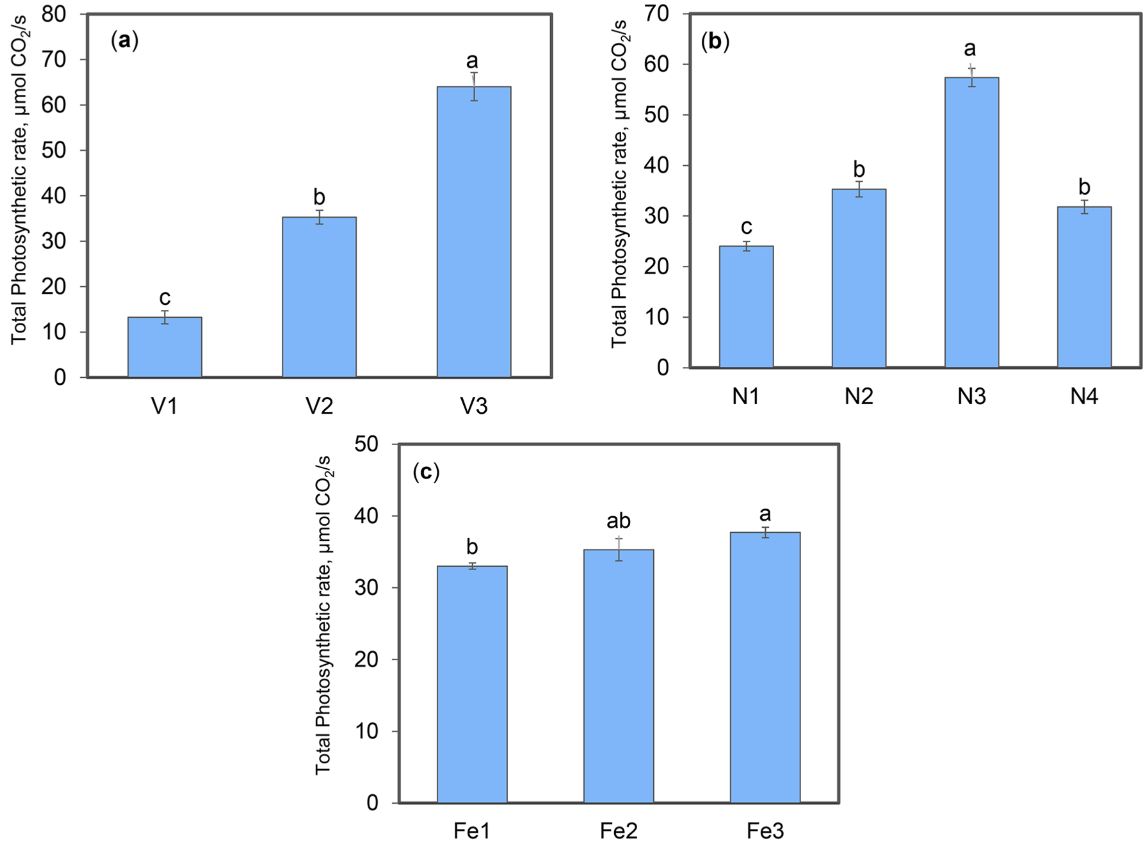

Fig. 5. The effect of (**a**) nutrient solution volume, (**b**) nitrate concentration, and (**c**) iron concentrations on the total photosynthetic rate (TPR). Values are means ± standard error ($n = 15$). The means followed by the same letter are not significantly different from each other, using LSD's multiple tests ($P < 0.05$).

the total cormlet yield. The number of large-sized cormlets accounted for 26% and 15% of the total cormlet yield in nutrient solution volume and nitrate treatment in a solution with EC 1.5 dS/m, respectively. In contrast, Salas et al.[35] only gained a 10% increase in the total number of large-sized cormlets by applying a nutrient solution with EC 3 dS/m.

Although the volume of nutrient solution and nitrate concentration affect the net photosynthetic performance of the saffron plant, iron had no clear effect on the net photosynthesis process. The highest transpiration and net photosynthesis values were achieved with a medium iron concentration (Fe2 = 25 µM). Applying a higher iron concentration (Fe3 = 30 µM) in the nutrient solution decreased the total weight of cormlets and large-sized cormlets, with no visible iron toxicity symptoms on leaves during the 5-month experimental period. Similar results have been reported by Adamski et al.[44] in sweet potato, a geophyte plant, where exposure to elevated iron concentrations (more than 4.5 mM Fe as Fe-EDTA) showed no toxicity symptoms. The normal concentration of Fe in plants ranges from 30 to 300 µg/g dry weight, depending on the plant species, physiological state, and growth conditions[45]. Although limited data exist on the optimum iron concentration for successful saffron production under hydroponic conditions, the decrease in functionality of some physiological responses including photosynthesis and growth inhibition may demonstrate the detrimental effects of excessive iron concentration[46]. High concentrations of Fe^{2+} ion may lead to the formation of reactive free radicals that oxidize and degrade proteins, lipids, carbohydrates, and DNA, potentially causing root cell destruction and ultimately cell death. The findings of Chatterjee et al.[46] indicate that the metabolic disorder observed in Fe-stressed potato leaves is also evident in the tubers, which may in turn decrease the number, size, and quality of tubers. Additionally, a significant decrease in the concentration of sugars, starch, and protein in potato tubers has been reported with increasing iron supply, indicating a disruption in carbohydrate and nitrogen metabolism.

The chemical analysis of cormlets provided a more in-depth understanding of how nutrients impact the photosynthesis process. Increasing nitrate concentration to 9800 µM (N3) while reducing iron concentration to 20 µM (Fe1) enhances nitrogen and iron uptake by cormlets, which are the two most important nutrients in photosynthesis and carbohydrate production[42]. These results are consistent with a recent study indicating that intercropping efficiency improves with iron foliar spraying, leading to significant enhancements in photosynthetic characteristics, related enzymes, nitrogen use efficiency, and crop yield[47]. Iron plays a vital role in nitrogen assimilation and translocation by regulating cofactor enzymes involved in nitrogen metabolism[48].

Among all nitrogen and iron levels, the highest iron accumulation (44 µg/g) was observed in cormlets produced under 9800 µM (N3), whereas the iron concentration of cormlets grown in 30 µM (Fe3) treatment

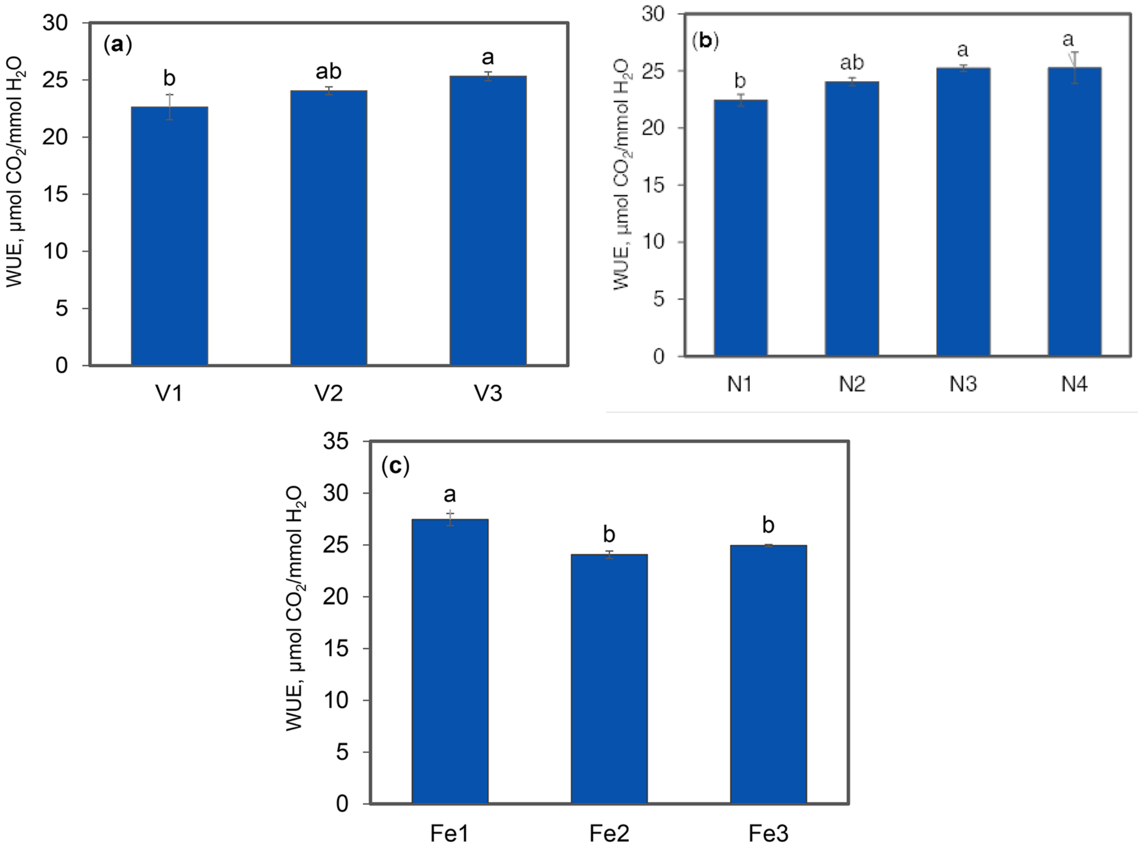

Fig. 6. The effect of (**a**) nutrient solution volume, (**b**) nitrate concentration, and (**c**) iron concentration on water use efficiency (WUE). Values with the same letter are not significantly different from each other, using LSD's multiple tests ($P < 0.05$).

(21 µg/g) was the lowest among different Fe levels (Table S6). The iron concentration in saffron corm has been reported to be around 40 µg/g[49]. In another case, a significant difference in Fe concentration between dormant (36 ± 0.3 µg/g) and waking (10.92 ± 0.8 µg/g) saffron corms has been observed[50]. While no visible symptom of Fe toxicity was observed in our experiment, the reason why cormlet production responds negatively to Fe concentrations, despite the rise in net and total photosynthetic rates with higher Fe concentration, remains unclear.

The efficiency of a hydroponic cultivation system can be evaluated by instantaneous water use efficiency (WUE) and nutrient solution productivity (NSP). A slight decline in WUE from the lowest (V1 = 75 mL/pot/w) to the intermediate (V2 = 150 mL/pot/w) nutrient solution volume suggests saffron's low water requirement, indicative of drought-tolerant plant species[51]. This could be attributed to decreased net photosynthetic rates (2.2 µmol CO_2/m^2/s), stomatal closure, and reduced water loss through transpiration. In addition, under water stress conditions supplying water from corm reservoirs helps the plant to alleviate low photosynthesis rate and cormlet growth in low water availability.

On the other hand, increasing nitrate concentration increased both net photosynthetic and transpiration rates, while not significantly impacting their ratio (WUE). Studies have reported contradictory effects of nitrogen supply on net photosynthetic rates and stomatal conductance. Some studies have shown that nitrogen supply can increase WUE through various concurrent physiological processes. These mechanisms involve maintaining a consistent net photosynthetic rate while decreasing transpiration[52], increasing the net photosynthetic rate by investing more nitrogen in the photosynthetic apparatus[53] without affecting stomatal conductance[54], or showing a moderate increase in photosynthetic rate alongside a slight decrease in stomatal conductance[55]. However, other studies have reported the conflicting effects of nitrogen supply on WUE[56], such as decreasing WUE or no significant effect[57]. In the case of the saffron plant, the physiological response of stomata conductance and photosynthesis rate to iron treatments caused a higher WUE at the lowest iron level due to reduced leaf transpiration. This finding is consistent with previous investigations[58] on the response to iron deficiency. Conversely[59], found increased stomatal opening and enhanced transpiration under insufficient iron concentrations.

In terms of nutrient solution productivity (NSP), increasing the nutrient solution volume caused an increase in production of large-sized cormlet but a decrease in production per unit of applied nutrient solution. This indicates that saffron has a low water requirement and can adapt well to water stress conditions. When nitrate concentration was increased to 9800 µM (N3), it led to an approximately 1.7-fold increase in NSP compared to

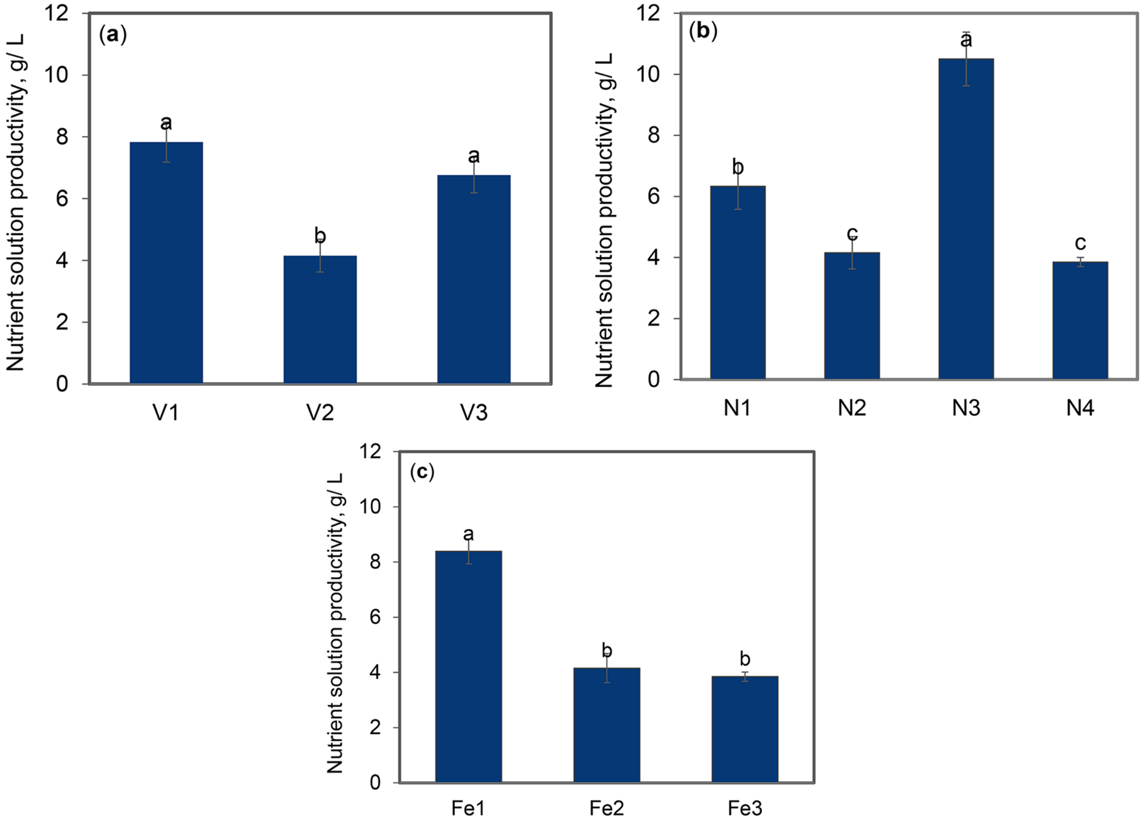

Fig. 7. Nutrient solution productivity (NSP) as a function of (**a**) nutrient solution volume, (**b**) nitrate concentration, and (**c**) iron concentration. Values with the same letter are not significantly different from each other, using LSD's multiple tests ($P < 0.05$).

the lowest nitrate concentration of 8100 µM (N1). However, when nitrate concentration was raised to 10,100 µM (N4), the weight of large-sized cormlets decreased, leading to a notable decline in productivity. These findings underscore the crucial role of proficient nitrogen management in enhancing saffron production. High iron concentration had a detrimental effect on the production of large-sized cormlets, leading to a 50% decrease in NSP at higher iron concentrations.

Even though both WUE and NSP showed similar trends with increasing the volume of nutrient solution, the highest nutrient solution volume (V3 = 225 mL/pot/w) resulted in a significant (50%) increase in the production of large-sized cormlets weighing above 8 g. Corm size plays a crucial role in the growth and yield of saffron. Generally, larger and more developed mother corms yield more flowers and larger daughter cormlets. Based on this, corms weighing over 8 g are considered the optimal corms in terms of quantitative and qualitative traits of saffron[60]. Compared with smaller corms, larger corms appear to have slightly earlier cell division and leaf growth development. The earlier the growth and subsequently the longer leaves result in greater photosynthetic products, followed by larger cormlet production.

The findings imply that while saffron can adapt well to suboptimal conditions, appropriate irrigation can enhance the quality of the next-generation cormlets. Furthermore, managing nutrient concentration can enhance the growth and production of cormlets. To maximize the economic value of the system, it is highly recommended to optimize nitrate (9800 µm) and iron (20 µm) concentrations along with a high nutrient solution volume (225 mL/pot/w).

Conclusions

The research findings underscore the crucial role of substrate composition and irrigation events in influencing the growth of saffron and the quantity and quality of cormlets produced. A substrate consisting of a low organic material and a high inorganic material (e.g., 15% cocopeat, 15% cocochips, and 70% perlite) can maintain an optimal balance of water and air in the root environment.

Increasing the volume of nutrient solution and the concentration of nitrate up to 9800 µM significantly increases the weight of cormlets, particularly large-sized cormlets. In contrast, increasing iron concentration negatively influences saffron performance, resulting in a decrease in the weight of both total and large-sized cormlets, while boron concentration did not affect cormlet production.

The experimental treatments did not affect the number of saffron leaves but significantly affected their weight. Notably, leaf weight positively correlated with the weight of total and large-sized corms in the experiments involving nutrient solution volume and nitrate concentration, while showing a negative correlation in the treatments involving iron concentration.

Increasing nitrate concentration in the nutrient solution led to an increased concentration of nitrogen and iron in cormlets while increasing iron concentration decreased their concentration. Additionally, increasing nutrient solution volume and nitrate concentration increases rates of total photosynthesis, demonstrating a positive correlation with the weight of total and large-sized corms, while in nitrate and iron treatments, the yield of cormlets was not affected by the net photosynthesis rate.

In summary, the findings of our study showed that optimizing nutrient solution volume and nitrate concentration, while considering iron levels, has the potential to significantly enhance saffron production in the coming years.

Materials and methods

Experimental conditions and plant materials

The experiments were carried out in a hydroponic greenhouse over the two growing seasons from 2018 to 2020. The saffron corms used in the experiments were obtained from a traditional saffron production region (landrace of Khorasan Province, Iran), which produces a dominant fraction of the world's saffron supply. To obtain a representative sample, only visually healthy corms with an average weight of 12–15 g were used for the experiments. Before planting, the corms were soaked in 1% Revoral fungicide solution for 20 min and then air dried.

Study design and treatments

The first year

In the first year of the experiment, the effects of phosphate and boron concentrations and the effects of irrigation events on cormlet production were studied by setting up a greenhouse pot experiment in a completely randomized design (CRD). The experimental treatments were three levels of phosphate concentration (P1 = 620, P2 = 800, and P3 = 980 µM) and three levels of boron concentration (B1 = 20, B2 = 25, and B3 = 30 µM), which were applied by using the six nutrient solutions with a predefined ionic composition as given in Table S1 in terms of µmol/L (for stoichiometry calculations) and in Table S2 in terms of mg/L (for more practical usage). Each of the six phosphate and boron treatments was applied three times, i.e., they were applied daily, twice weekly, and weekly, resulting in a total of 18 treatments. Each treatment was replicated five times, for a total of 90 pots.

The pots used in the experiment were 20 × 25 cm in size (width by height) and were filled with a mixture of medium-sized cocopeat (70%) and medium-sized perlite (30%). Three corms were planted in each pot at a depth of 15 cm. Since saffron is a synanthous plant, it flowers before the leaves emerge. During this phase, the plants lack roots and require only a moist medium for initial growth. Therefore, prior to flowering, the pots were irrigated weekly with tap water from the greenhouse to provide initial moisture for the flowering process. The pH of the tap water was adjusted to approximately 7 by adding a certain amount of sulfuric acid. The ionic composition of the tap water can be found in Table S1 in terms of µmol/L and in Table S2 in terms of mg/L.

After cease of flowering, the experimental pots were irrigated with the six nutrient solutions, described above. The initial pH of the nutrient solutions was 5.8 ± 0.2, and their electrical conductivity (EC) was 1.1 dS/m. The treatments commenced at the end of the flowering period in November and continued until leaf senescence. Finally, the number and weight of cormlets were determined.

The second year

In the second year of the experiment, some changes were made based on the results obtained in the first year to improve the experimental setup. Firstly, taller pots measuring 20 × 40 cm (width by height) were used to facilitate greater root development. Secondly, the pots were filled with a mixture of cocochips (15%), cocopeat (15%), and perlite (70%) to have a lower water-holding capacity and higher aeration capacity than the first-year experiment. Thirdly, the pots were irrigated only weekly. Fourthly, a broader array of nutrients and their concentrations were chosen to assess their impacts on the production of cormlets. Similar to the first year, three corms were planted at a depth of 15 cm in each pot.

Accordingly, the effects of four different treatments, including the concentrations of nitrate, iron, and boron, as well as the volume of nutrient solution, on the growth of saffron and cormlet production were studied using a completely randomized design (CRD). The experimental treatments consisted of four nitrate concentrations (N1 = 8100, N2 = 9300, N3 = 9800, and N4 = 10100 µM), three iron concentrations (Fe1 = 20, Fe2 = 25, and Fe3 = 30 µM), and four boron concentrations (B1 = 18, B2 = 25, B3 = 32, and B4 = 46 µM). Additionally, to evaluate the effect of nutrient solution volume, three volumes of N2 nutrient solution (V1 = 75, V2 = 150, and V3 = 225 mL/pot/w) were applied. All 14 treatments were applied with a predefined ionic composition and were replicated five times. The specific ionic composition of the nutrient solutions is given in Table S3 in terms of µmol/L and in Table S4 in terms of mg/L.

Each pot in every treatment, except for those in V1 and V3 treatments, received 150 mL of their specific nutrient solution weekly. The pots in the V1 and V3 treatments received 75 mL and 225 mL of nutrient solution weekly, respectively. Every five pots, which were treated similarly, were irrigated independently through an automated tape-irrigation system (one hole per pot) and this continued until leaf senescence, almost at the end of April. Similar to the first year, the initial pH of the nutrient solutions was approximately 5.8, however, a higher EC, 1.5 dS/m, was used.

As per the data presented in Tables S1 and S3, the concentration of all ions, excluding the ones that were being studied, remained constant in all nutrient solutions. However, to maintain a consistent total ionic concentration (i.e., EC = 1.1 and 1.5 dS/m, in the first and second year, respectively) in all the applied treatments in the first year, the nitrate concentration was decreased in proportion to the increase in phosphate concentration (the experimental treatment). Similarly, in the second year, the sulfate concentration was decreased in line with the rise in nitrate concentration (the experimental treatment). The changes in iron and boron concentrations (the experimental treatments) had a negligible effect on the overall ionic composition of the solutions. The nutrient solutions were made using tap water and commercial fertilizers. The ionic composition of the tap water and nutrient solutions are shown in Tables S1 to S4.

Moisture characteristics of the substrate

In order to optimize the substrate composition, the moisture characteristics of various substrates with varying ratios of cocopeat to perlite was determined using a weighting method. A pot was filled with a known ratio of cocopeat to perlite, and its weight was measured. It was then placed in water for 24 h to ensure complete saturation. Afterward, it was weighed again and left for 12 h to eliminate excess water, following which its weight was measured once more. The difference between the first and second weights was considered representative of saturated moisture, while the difference between the first and third weights was considered representative of the water holding capacity[61,62]. The volumetric percentage of each moisture index calculated by dividing each parameter by the volume of the substrates.

Gas exchange parameters

The photosynthetic rate of plants was measured in March when leaf and root development had ceased and the saffron cormlets were in the growth phase[10]. A portable system called the LICOR-6400 (LI-COR Biosciences, Lincoln, NE), equipped with a leaf chamber and a light source emitting a mixture of blue and red lights, was used to measure photosynthesis. The net photosynthetic rate (*Pn*) and transpiration rate (*Tr*) were recorded under the carbon dioxide (CO_2) concentration in the inflow air of ≈ 350 μmol/mol and leaf temperature of 27 ± 2°C. The measurements were performed on the leaves of three randomly selected pots in each treatment. The total photosynthetic rate was calculated taking into account the net photosynthetic rate and the leaf area (LA), which was estimated by multiplying the length by the average width of the leaves using graph paper. The instantaneous water use efficiency (WUE) was also calculated in all treatments by Eq. 1[63]:

$$WUE = P_n/T_r \tag{1}$$

in which P_n is the net photosynthetic rate and, T_r is the transpiration rate.

Harvest and analysis

After about five months of being exposed to the treatments, the 6-month-old saffron plants were harvested in April 2019 and April 2020. In both years, the number and total weight of produced cormlets and the large-sized cormlets (per corm) were measured. In addition, in the second year, the shoots and cormlets were separated, and several growth parameters, including the LA, and the number of fresh leaves per corm were measured. The cormlets were rinsed in tap water, washed with deionized water, and dried with tissue paper. They were then weighed and oven-dried at 70°C for 48 h until a constant weight was achieved. After determining the dry mass, the plant samples were ground using a stainless-steel mill for elemental analysis. To calculate the productivity of each treatment, the fresh weight of produced large-sized cormlets per unit of applied nutrient solution at every stage of plant growth (g/L) was measured.

Tissue ion concentration

Dried powder samples of the second year were digested with a mixture of concentrated sulfuric acid and salicylic acid and left overnight. The obtained dark-colored materials were heated for one hour at 180 °C using a heating block. The obtained extracts were cooled and 33% hydrogen peroxide was carefully added and the mixture was heated again at 280 °C for 5 to 10 min to see the white fumes. Hydrogen peroxide addition was repeated several times until a clear sample solution was produced[64]. The extracts were cooled, filtered, and diluted to a volume of 25 mL with deionized water and stored in plastic vials until analyzed. The concentration of Fe and N was measured by flame atomic absorption spectroscopy (FAAS) and micro-Kjeldahl methods, respectively.

Statistical analysis

Before the analysis of variance (ANOVA), residuals were tested for normality using the Proc UNIVARIATE in SAS 9.3 (SAS Institute, Cary, NC, USA). ANOVA was done using the Proc GLM (general linear model) in SAS, followed by the LSD (least significant difference) test as a post hoc ($p \leq 0.05$).

Use of AI tools

The authors declare that they have not used any generative artificial intelligence for the writing of this manuscript, nor for the creation of images, except using simple tools for identifying grammatical mistakes.

Data availability

Experimental data will be available on request (from: Rasoul Rahnemaie).

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Declarations

Competing interests
The authors declare no competing interests.

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